Failure Analysis Report

FAIL-0433 — Synchronization Failure

1. Failure Summary

Failure ID	FAIL-0433
Failure Type	Synchronization Failure
Severity	Minor
Component	Cache
Application	APP-007 — InternetBrowser
Build Version	4.0.6
Timestamp	2023-09-23 22:16:23
Environment	ENV-006 — Cloud Test Env
OS Version	Ubuntu 22.04
Network	Ethernet
Resolution	In Progress

2. Test Case Reference

Test Case ID	TC-0033
Title	Ensure that configuration reload does not exceed the CPU ceiling (80%) threshold...
Execution ID	EXEC-02105
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0433 — Reliability — Payment Processing

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete session token refresh within the defined SLA.
Observed Behaviour	The system exhibited synchronization failure when the network connection is interrupted mid-flow.

4. Error Details

Error Message:

```
SyncError: conflict detected on entity 'ENTITY-2010'; last_modified timestamps diverge by 97s.
```

Stack Trace / Log Excerpt:

```
com.samsung.browser.SyncEngine.execute(line 231)
```

Log Reference	LOG-99573

5. Executed Steps at Time of Failure

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1	Record response time using the built-in performance monitor.	A success toast message is displayed.
2	Initiate device pairing by pressing the 'Pay' button.	The UI transitions to the next screen within 3 seconds.
3	Disconnect the headset and wait 4 seconds.	The UI transitions to the next screen within 3 seconds.
4	Assert that database record for entity 'ENT-310' has status='FAIL...	The device reconnects automatically within 10 seconds.

6. Root Cause Analysis

Race condition in the connection manager when two threads attempt simultaneous reconnection.

7. Linked Defect

Defect ID	BUG-0258
Title	Synchronization Failure in Cache: SyncError: conflict detected on entity 'ENTITY...
Severity	Minor
Priority	Medium
Status	CLOSED
Root Cause	Race condition in the connection manager when two threads attempt simultaneous r...
Resolution	Added key rotation broadcast to all replica nodes; integration test verifies pro...
Assigned To	ENG-011

8. Historical Similar Failures

HIST-0172	FAIL-0173	Synchronization Failure	5.4.7	2022-01-09	15	Wont Fix
HIST-0251	FAIL-0252	Synchronization Failure	5.2.3	2021-10-08	5	Resolved
HIST-0432	FAIL-0433	Synchronization Failure	4.2.1	2021-02-07	5	Wont Fix

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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